

Assignment 1

Explore following 4 websites and answer the questions

- bitcoin.org
- Blockchain.com
- <https://coinmarketcap.com/>
- <https://www.investopedia.com/>

One-Word Answerable Questions from bitcoin.org

Bitcoin Basics

1. Bitcoin was introduced by whom?
Satoshi
2. Bitcoin transactions are recorded on which data structure?
Blockchain
3. Bitcoin uses which consensus mechanism?
POW
4. The smallest unit of Bitcoin is called?
Satoshi
5. Bitcoin is a **peer-to-peer** type of what system?
Currency

Wallets

6. A wallet stores the user's **Private** key.
7. The recovery phrase used in wallets is called a **Seed** phrase.
8. Hardware wallets store keys **Offline** (online/offline).
9. Bitcoin addresses are derived from **Public** keys.
10. What is required to sign a transaction? (public/private)
Private

Security

11. Bitcoin prevents double-spending using Consensus.
12. Bitcoin is secured using Asymmetric cryptography (asymmetric/symmetric).
13. To run a Bitcoin node, the software required is Bitcoin Core.
14. The process of verifying transactions is called Mining.
15. The Bitcoin network is described as Decentralized (centralized/decentralized).

Mining

16. Miners solve Cryptographic puzzles to add blocks.
17. Mining rewards a block with newly created Bitcoin.
18. The block time target for Bitcoin is Ten minutes.
19. Mining difficulty adjusts every 2016 blocks.
20. Miners compete to find a valid block Hash (hash/signature).

Blockchain Structure

21. Each block contains a Merkle root.
22. Blocks are linked using the previous block's Hash.
23. A Bitcoin transaction contains inputs and Outputs.
24. Unspent transaction outputs are called UTXOs.
25. The first-ever Bitcoin block is called the Genesis block.

Network

26. A full node stores the entire blockchain Data.
27. Bitcoin uses which protocol for communication? (TCP/P2P) **P2P**
28. A light wallet uses SPV mode (SPV/full).
29. Nodes relay information across a distributed network.

30. The maximum Bitcoin supply is 21 million.

Task B1: Explore a Block

Note the following from the **latest block**:

- Block height: 1
- Block hash:
000000000000000000021c388077cb64df868708b37ca088ff4066943cb1863f
- Timestamp: Mined on August 03, 2026 11:44:14
- Number of transactions: 6,265
- Miner/pool name: Volatility

Task B2: Explore a Transaction

Open any one Bitcoin transaction.

10. Write the TXID of the transaction.
8363ce49e0bcede985a9a3b60e9bb3929994c31637c26eb457f44f7cd464b0fe
11. Number of inputs: 1
12. Number of outputs: 2
13. Transaction fee: 0.00000049 BTC
\$0.03
14. How many confirmations does it have? 2

Task B3: Explore an Address

15. Search for any Bitcoin address shown in the block.
bc1qngxepmspp6es79703f820wdvsgta0eptx6q
16. Record:
- Total BTC received: 0.02036552 BTC \$1277.87
 - Total BTC sent: -0.00015520 BTC \$-9.74
 - Current balance: 0.02021032 BTC \$1268.13

What is the role of a full node?
Storing entire history of the block chain

List any two developer resources available on bitcoin.org.
Charts and Tools
Lightning Network

Explore : <https://coinmarketcap.com/>

Task C1: Market Information

- Current price of BTC:
62,712.00 USD
- 24-hour % change: -2.71%
- Market capitalization: \$1.274 trillion USD
- Circulating supply: 20.06 million BTC
- Total/maximum supply: 20.06 million BTC / 21 million BTC
- **Task C2: Historical Data**
- Bitcoin price one year ago (approximately): \$66,800 USD (around early August 2025).
- Highest price in the last 30 days: \$66,934 USD
- Lowest price in the last 30 days: \$59,713 USD

Explore <https://www.investopedia.com/>

Task D: Understanding the Market

Why is Bitcoin considered volatile?

Bitcoin is considered volatile because its price can change rapidly over short periods due to factors such as market demand, investor sentiment, government regulations, macroeconomic news, and large transactions by major holders.

Two investment risks mentioned:

High price volatility – The value of Bitcoin can rise or fall significantly in a short time, leading to substantial gains or losses.

Regulatory risk – Changes in government regulations or legal restrictions can negatively affect Bitcoin's price and adoption.

Part E: Reflection & Analysis

Which website gave you the most clear understanding? Why?

CoinMarketCap gave me the clearest understanding because it presents key information such as price, market capitalization, trading volume, supply, and historical data in a simple, well-organized dashboard. It also updates data in real time, making it easy to understand Bitcoin's current market performance.

In one paragraph, explain how blockchain ensures trust without a central authority.

Blockchain ensures trust without a central authority by storing transactions on a distributed network of computers instead of a single server. Every transaction is verified by network participants through a consensus mechanism before it is added to the blockchain. Once recorded, transactions are encrypted and linked to previous blocks, making them extremely difficult to alter or delete. Because every participant has a copy of the blockchain, anyone can verify transactions, creating transparency, security, and trust without relying on a bank or other central institution.

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